

RUTLAND SOUTHERN VERMONT, VT, USA

WMO: 725165

Lat: 43.533N

Lon: 72.950W

Elev: 784

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-0.3	-20.2	1.9	2.6	-14.8	2.5	5.2	26.6	34.3	24.1	32.8	5.0	130	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.1	70.5	82.3	68.7	80.9	67.5	73.2	81.9	71.4	79.6	69.6	77.2	7.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	114.9	78.3	68.4	107.6	76.6	66.3	99.8	74.8	37.2	81.6	35.6	79.7	34.2	78.3	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.2	18.6	16.5	DB	-14.0	89.9	6.1	2.6	-18.4	91.8	-22.0	93.3	-25.4	94.7	-29.9	96.6	(4)
			WB	-14.5	76.0	5.7	2.0	-18.6	77.5	-21.9	78.7	-25.2	79.8	-29.3	81.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	20.7	23.6	31.6	43.9	55.7	64.0	68.7	66.7	60.0	48.6	37.7	27.5
	DBStd	18.85	12.90	10.99	10.38	8.89	7.79	6.42	5.25	5.39	7.27	8.15	9.38	10.22
	HDD50	3693	910	739	576	225	30	0	0	0	6	128	381	698
	HDD65	7344	1374	1159	1034	634	305	93	23	45	185	511	819	1162
	CDD50	2177	1	0	6	43	206	421	579	518	306	83	13	1
	CDD65	354	0	0	0	2	17	64	137	98	34	2	0	0
	CDH74	2798	0	0	6	45	196	565	1070	657	248	11	0	0
	CDH80	598	0	0	1	8	37	129	254	120	49	0	0	0
Wind	WSAvg	7.0	7.5	7.7	7.9	8.4	7.4	6.5	5.8	5.6	6.2	7.0	7.5	7.3
Precipitation	PrecAvg	45.2	3.2	2.6	3.4	3.5	3.7	4.4	4.7	4.1	3.8	4.4	3.7	3.7
	PrecMax	53.7	6.2	6.3	6.3	6.9	6.3	10.3	8.8	7.4	8.5	10.5	5.5	6.3
	PrecMin	33.2	1.1	1.1	1.5	0.8	1.2	1.3	1.9	1.3	1.9	0.5	2.3	1.6
	PrecStd	6.1	1.3	1.1	1.2	1.4	1.5	2.0	1.9	1.5	1.5	2.6	1.0	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	56.4	69.8	80.7	84.0	88.3	89.7	87.9	85.8	75.2	66.5	57.1
		MCWB	52.8	47.7	55.9	60.6	66.4	72.2	74.4	71.9	70.3	62.5	57.3	50.1
	2%	DB	48.4	48.8	58.5	72.2	79.9	83.7	85.2	82.4	80.8	71.2	62.5	51.6
		MCWB	44.1	42.7	48.4	55.2	62.6	69.8	71.4	69.3	67.5	61.6	54.0	46.6
	5%	DB	42.9	43.5	52.2	65.6	74.8	80.9	82.3	81.0	76.4	66.1	56.9	45.9
		MCWB	38.1	37.6	43.7	52.6	60.3	66.7	69.8	68.4	66.0	58.1	50.4	42.1
	10%	DB	37.1	38.6	46.1	60.6	71.0	77.1	80.7	78.5	72.6	62.7	52.4	41.2
		MCWB	33.6	34.0	39.5	49.3	58.3	64.6	68.4	66.8	63.8	55.1	46.5	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	50.3	56.6	63.2	68.9	74.2	76.4	74.1	72.7	67.0	59.8	51.5
		MCDB	57.6	54.7	66.7	75.0	80.8	84.7	85.9	82.4	80.6	71.8	64.6	56.0
	2%	WB	45.2	43.1	50.1	58.0	66.0	71.5	73.9	71.9	69.8	62.5	55.1	46.7
		MCDB	47.9	48.0	56.7	69.2	75.5	80.4	82.7	80.0	77.3	68.3	60.2	49.4
	5%	WB	38.4	38.1	44.4	53.8	63.0	69.1	71.8	70.2	67.5	59.2	51.4	42.2
		MCDB	42.0	42.5	50.4	64.0	71.5	77.1	80.1	77.7	74.3	65.5	56.1	46.1
	10%	WB	34.2	34.6	40.1	49.6	60.1	66.7	70.0	68.4	65.3	55.5	46.6	37.5
		MCDB	37.5	38.7	46.2	58.8	68.6	74.2	77.5	75.5	71.2	61.6	51.1	40.7
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.7	16.9	19.2	20.2	19.3	19.1	19.1	19.1	17.0	14.7	13.6
		MCDBR	19.5	20.5	25.3	29.5	27.1	23.9	21.5	20.9	22.7	23.8	21.1	18.0
		MCWBR	17.3	16.1	16.9	17.1	14.4	12.6	10.9	11.0	12.4	14.9	16.1	15.9
	5% WB	MCDBR	18.7	19.3	22.6	25.8	22.6	20.2	18.9	18.2	19.2	20.5	18.6	17.7
		MCWBR	17.2	16.2	16.9	16.8	13.6	11.6	10.4	10.3	11.5	14.5	16.2	16.4
Clear-Sky Solar Irradiance	taub		0.290	0.297	0.322	0.346	0.379	0.422	0.438	0.414	0.365	0.337	0.306	0.283
	taud		2.361	2.319	2.384	2.414	2.363	2.255	2.216	2.271	2.398	2.487	2.519	2.463
	Ebn at Noon		266	285	292	292	283	270	264	267	273	268	262	260
	Edh at Noon		25	31	33	35	38	43	44	40	33	26	22	21
All-Sky Solar Radiation	RadAvg		491	769	1133	1416	1600	1727	1783	1611	1303	800	535	379
	RadStd		46	67	81	166	183	151	131	85	97	86	59	40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A

Nomenclature: See separate page